Exact multiplicity and solver controls

2026-11-09. All 16 registered msolve 0.6.5 calls completed with exit 0 and passed the frozen native checks. Lean proves an actual two-variable ideal has one distinct F5 point but quotient length 2. This refutes equating support count with algebraic multiplicity; it does not refute Diem's bound. Claim PCC-20261109-ee57d0, unsigned. [pcc:unverified]

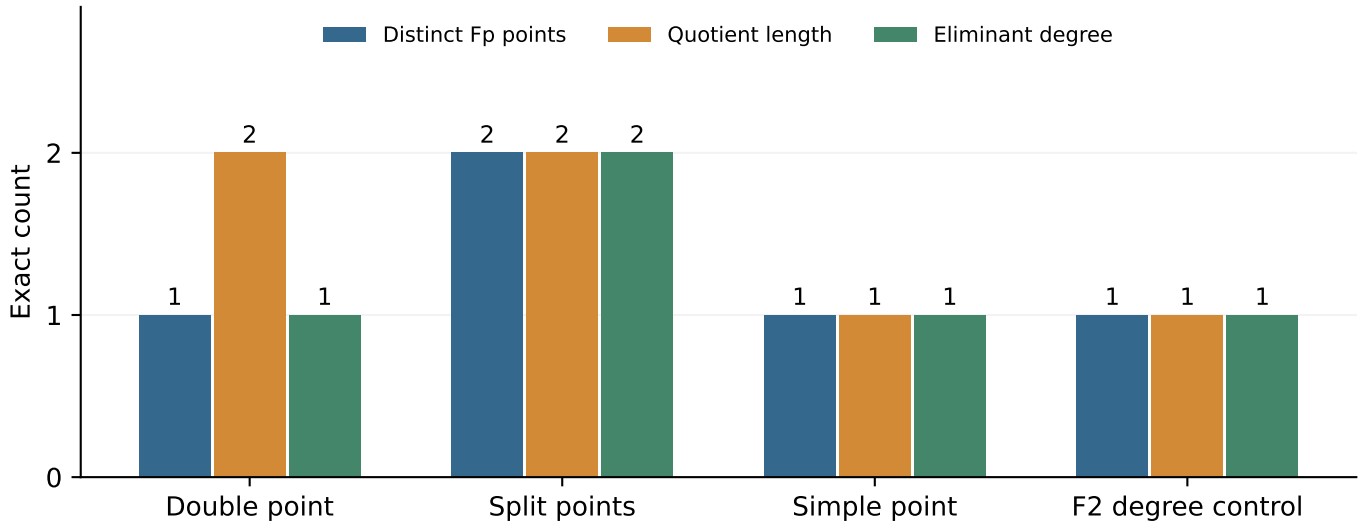

Figure 1: Recorded exact counts. Each fixture has two repetitions in each mode; both repetitions agree. No sampling interval or accepted clock metric. Run solver-multiplicity-20261109-v1.

The formal family is $I=(xy,x+y-c)$. Lean constructs $K[x,y]/I \simeq_a [K]\ K[t]/(t^2-ct)$, proves dimension2, proves equality with the triangular ideal $(x+y-c,y^2-cy)$, and classifies every field solution. At $c=0$, the unique point has total multiplicity 2. At $c\neq 0$, the two points are distinct. Closed declaration: BivariateMultiplicityAudit.support_is_not_multiplicity. Expanded transitive gate: 93 declarations, 0 failing, 0 trusted-native exceptions; no custom axiom, sorry or native_decide. [pcc:unverified]

msolve prints qdim 2 and eliminant degree 1 for the double point, 2 and 2 for split points, 1 and 1 for the simple point and F2 degree control. Independent native analysis checks normalized bases, input reductions, S-polynomials, finite standard-monomial counts and every base-field point. The degree control $\{x^2+x,y^2+y,xy+x+y\}$ returns $\{x,y\}$; both GB runs select pair degree 3. That logged degree is exploratory; the prior Lean lower/upper certificates are separate mathematical statements. [pcc:unverified]

Flow: 16 planned, 16 attempted, 16 completed; 0 not-completed, 0 interrupted, 0 resumed, 0 excluded. Four fixtures × two modes × two repeats. Exact sparse linear algebra (-l2), one thread, no saturation, 30-second timeout per call. msolve source commit 5e72b9d9; binary SHA256 prefix 390a41bd8d8c. Native analysis and frozen protocol committed before calls in f0bea543. Protocol hash prefix cff556238d40; full identities are in protocol.json and provenance.json. The canonical export and every referenced file hash validate. [pcc:unverified]

Held-out c=2 preflight was seen before freeze and only calibrates format and budget. Its wrong-constant/zero-polynomial controls pass. No A/A variance, null population, significance or confidence claim: these are exact known answers. Default genericity can add variables and use time-seeded random forms; raw parameters and containing epoch intervals are retained. Counts, not parameter bytes, are compared. One shared local host, zero paid spend, approved local checkpoints, no accepted timing or independent-host claim. [pcc:unverified]

Sources: pinned msolve main.c and msolve.c:123–181 document distinct qdim and parameter fields; upstream nonradical_shape-31 already prints 11 and 8. Mathlib AdjoinRoot gives the quotient basis machinery. Diem2011's general $n!d^n$ bound, C-to-F4Run refinement and the remaining14-item/paper backlog remain open. Raw data:../../evidence/runs; exact checks:../../evidence/analysis.jsonl; schema/hash-validated export:../../export.json; source/build/dependencies/host: ../../provenance.json. [pcc:unverified]